

Data Centers & Energy: The Infrastructure Behind AI

Data centers are the beating heart of artificial intelligence. Every query to a large language model, every training run of a neural network, every AI-generated image is computed in vast warehouses of servers that consume enormous amounts of electricity and water. This sheet examines key data center sites and their environmental impacts.

Ashburn, Virginia — Data Center Alley

The World's Largest Concentration of Data Centers

Loudoun County, Virginia — centered on the town of Ashburn — hosts the densest cluster of data centers on Earth. Often called "Data Center Alley," this corridor houses over 300 data centers operated by Amazon Web Services, Microsoft Azure, Google, Meta, Equinix, Digital Realty, and dozens of others. An estimated 70% of the world's internet traffic passes through this region daily.

The concentration began in the 1990s due to proximity to MAE-East (one of the original internet exchange points), abundant fiber-optic infrastructure, and business-friendly Virginia policies including tax incentives for data center construction.

Energy and Grid Impact

The explosive growth of data centers — accelerated by AI demand — is straining Virginia's electrical grid. Dominion Energy, the regional utility, has reported that data center electricity demand in Northern Virginia could reach 30+ GW by 2040, up from approximately 3.5 GW in 2022. This is prompting construction of new natural gas plants and controversial proposals for small modular nuclear reactors.

Land Use and Community Impact

Residential communities increasingly oppose new data center construction due to noise (backup generators, cooling systems), light pollution, and the transformation of rural landscapes. In 2023, Loudoun County introduced stricter zoning rules after significant community pushback.

Key Statistics

- Loudoun County hosts over 300 data centers, more than any other jurisdiction globally (Loudoun County EDA, 2023).
- Data centers consumed approximately 25% of Dominion Energy's Virginia load in 2023 (Dominion Energy).
- Northern Virginia data center market: approximately 2,800 MW of IT load capacity (CBRE, 2023).
- Tax revenue from data centers to Loudoun County: approximately \$600 million/year.
- Projected electricity demand growth: 5–7 GW of new data center load expected by 2030 in Virginia alone.
- An estimated 70% of global internet traffic transits through Ashburn (various industry estimates).

Sources: Loudoun County Economic Development Authority (2023); Dominion Energy Integrated Resource Plan (2023); CBRE North America Data Center Report (2023); Washington Post reporting on data center growth (2023).

The Dalles, Oregon — Google's Water Problem

A Small Town, a Giant Thirst

The Dalles, a town of approximately 15,000 people on the Columbia River in Oregon, hosts one of Google's largest data center campuses. The facility was built starting in 2006, attracted by cheap hydroelectric power from the Bonneville Power Administration and a cool, dry climate.

Water Consumption Controversy

In 2022, Oregon state records revealed that Google's The Dalles campus consumed approximately 12.7 billion liters (3.36 billion gallons) of water in 2021 — more than a quarter of the city's total water usage. Much of this water is used in evaporative cooling systems. During heat waves, consumption spikes dramatically.

Google had initially fought to keep its water usage data confidential, claiming it was a trade secret. Public pressure and Oregon state transparency requirements eventually forced disclosure.

The situation is particularly contentious because The Dalles draws its water from the Columbia River watershed during a period of increasing drought stress in the Pacific Northwest, with salmon populations already under severe pressure.

Key Statistics

- Google's The Dalles campus used approximately 12.7 billion liters of water in 2021 (Oregon Water Resources Department).
- This represented over 25% of The Dalles' total municipal water usage.
- Google globally consumed approximately 21.2 billion liters of water in 2022, a 20% increase from 2021 (Google Environmental Report, 2023).
- The company acknowledged that AI workloads significantly increase water consumption per query.
- Google proposed a new expansion that could increase water usage by an additional 3.8 billion liters annually.
- Globally, data centers consumed an estimated 1.7 billion liters of water per day in 2022 (IEA, 2023).

Sources: Oregon Water Resources Department records; Google Environmental Report (2023); The Oregonian investigative reporting (2022); IEA, *Data Centres and Data Transmission Networks* (2023); Li, P. et al., "Making AI Less Thirsty", University of California Riverside (2023).

Dublin, Ireland — The Moratorium

Europe's Data Center Capital

Ireland, and Dublin in particular, has become Europe's largest data center market. Attracted by low corporate tax rates, EU market access, cool climate, and English-speaking workforce, tech giants have built massive campuses across the Dublin region. By 2022, data centers accounted for approximately 18% of Ireland's total electricity consumption — and this figure was rising rapidly.

The Grid Crisis

In November 2021, EirGrid (Ireland's grid operator) issued an unprecedented statement: the Dublin region's electricity supply was at risk of shortfall. The explosive growth of data centers had outpaced grid infrastructure investment. In response:

- **EirGrid imposed a de facto moratorium** on new data center grid connections in the Dublin region.
- The Commission for Regulation of Utilities (CRU) introduced new rules requiring data centers to provide on-site backup generation that could also support the grid during peak demand.
- Several planned data center projects were delayed or relocated.

The situation highlighted a fundamental tension: nations competing to attract tech investment were discovering that data centers could consume grid capacity faster than it could be built, particularly when renewable energy targets required retiring fossil fuel plants.

Key Statistics

- Data centers consumed approximately 18% of Ireland's total electricity in 2022, up from 5% in 2015 (EirGrid, 2023).
- Ireland hosts approximately 82 operational data centers, with 30+ under construction or planned (Host in Ireland, 2023).
- Projected data center electricity demand: up to 30% of Ireland's total by 2028 if unchecked (EirGrid).
- EirGrid's moratorium affected over 30 planned data center grid connections in the Dublin area.
- Ireland's entire national electricity consumption in 2022: approximately 32 TWh; data centers: approximately 5.8 TWh.
- Meta's data center in Cloness, County Meath, has a maximum power draw of 200+ MW.

Sources: EirGrid, *All-Island Generation Capacity Statement* (2023); CRU Data Centre Policy Statement (2022); Host in Ireland annual report (2023); Irish Times and RTÉ reporting.

Iowa — GPT-3 Training

The Energy Cost of a Language Model

OpenAI's GPT-3, one of the landmark large language models, was trained in Microsoft Azure data centers in Iowa. The choice of Iowa was driven by access to relatively cheap electricity (a mix of wind and natural gas) and Microsoft's expanding data center presence in the state (West Des Moines campus).

Training Energy and Water

Training GPT-3 (175 billion parameters) was estimated to consume approximately 1,287 MWh of electricity, generating an estimated 502 tonnes of CO₂ equivalent. Researchers at the University of California Riverside later estimated that GPT-3 training consumed approximately 700,000 liters of water for cooling alone.

The subsequent GPT-4 model, while details remain less public, is estimated to have required 5–10x the compute of GPT-3. Industry estimates suggest GPT-4 training consumed on the order of 50,000–100,000 MWh of electricity.

Inference: The Ongoing Cost

While training captures headlines, inference (the ongoing cost of responding to user queries) now dominates total AI energy consumption. A single ChatGPT query is estimated to consume approximately 10x the electricity of a Google search query. With hundreds of millions of daily users, the aggregate inference cost is enormous and growing.

Key Statistics

- GPT-3 training: approximately 1,287 MWh of electricity, approximately 502 tonnes CO₂e (Patterson et al., 2021).

- GPT-3 training water consumption: approximately 700,000 liters (Li et al., 2023).
- GPT-4 training estimated at 5–10x the compute of GPT-3 (various analyst estimates).
- A ChatGPT query consumes approximately 0.001–0.01 kWh, roughly 10x a Google search (IEA, 2024).
- Microsoft's Iowa data center water consumption increased 34% from 2021 to 2022 (Microsoft Environmental Report, 2023).
- Global AI electricity consumption is projected to reach 85–134 TWh annually by 2027 (IEA, 2024).

Sources: Patterson, D. et al., "Carbon Emissions and Large Neural Network Training", arXiv (2021); Li, P. et al., "Making AI Less Thirsty", University of California Riverside (2023); IEA, *Electricity 2024* (2024); Microsoft Environmental Sustainability Report (2023).

Lulea, Sweden — A Best-Practice Example

Facebook's Arctic Data Center

In 2013, Meta (then Facebook) opened a data center in Lulea, a city in northern Sweden located just 100 km south of the Arctic Circle. The facility was designed as a showcase for sustainable data center operations.

Why Lulea Works

Several factors make Lulea exceptionally favorable:

- **Climate:** Average annual temperature of approximately 2 degrees C. For most of the year, outside air can be used directly for cooling (free cooling), eliminating the need for energy-intensive mechanical chillers.
- **Renewable energy:** Sweden's electricity grid is approximately 98% fossil-free, dominated by hydroelectric and nuclear power. The Lulea data center runs on 100% renewable energy from local hydroelectric sources.
- **Waste heat recovery:** Excess heat from the data center is captured and fed into Lulea's district heating network, warming homes and businesses in the city.

Power Usage Effectiveness (PUE)

The industry-standard metric for data center energy efficiency is PUE (Power Usage Effectiveness), defined as total facility energy divided by IT equipment energy. A PUE of 1.0 would mean all energy goes directly to computing; the global average is approximately 1.55–1.60.

Meta's Lulea facility achieves a PUE of approximately 1.07–1.10, among the lowest in the world. This means only 7–10% of energy is "overhead" (cooling, lighting, etc.), compared to 40–60% overhead in older or poorly designed facilities.

Limitations of the Model

While Lulea represents best practice, its model has inherent limits:

- Not all data centers can be located near the Arctic Circle; latency requirements often demand proximity to users.
- Even with a PUE of 1.07, the facility still consumes hundreds of megawatts of electricity.
- Sweden's grid, while clean, is facing increasing demand from industry (steel, batteries) and data centers, raising questions about long-term capacity.

Key Statistics

- PUE of approximately 1.07–1.10, compared to global average of approximately 1.55–1.60 (Meta, 2023; IEA, 2023).
- Powered by 100% renewable energy (hydroelectric).
- Waste heat recovery supplies approximately 25% of Lulea's district heating needs (Meta, 2023).
- Cooling relies entirely on outside air for most of the year (free cooling).
- Average annual temperature in Lulea: approximately 2 degrees C.
- Global data center electricity consumption: approximately 460 TWh in 2022, 1–1.3% of global electricity demand (IEA, 2023).

Sources: Meta Sustainability Report (2023); IEA, *Data Centres and Data Transmission Networks* (2023); Lulea municipality energy reports; Uptime Institute Global Data Center Survey (2023).

Conclusion

Data centers represent the operational phase of AI's material footprint — the ongoing, growing demand for electricity and water that persists long after chips are manufactured. The contrast between sites like Lulea (designed for efficiency in a favorable climate with clean energy) and the strain on grids in Virginia, Ireland, and elsewhere illustrates that location, design, and energy sources matter enormously. Yet even the best-designed data center cannot eliminate AI's fundamental dependence on physical resources. As AI workloads grow exponentially, efficiency gains risk being overwhelmed by sheer scale — a dynamic known as the Jevons paradox.